MIL-E-1/985B
15 July 1977
SUPERSEDING
MIL-E-1/985A
30 March 1970

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5U4GB

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Rectifier, full wave, high vacuum

Outline	---	12-16
Base	---	B5-121
Envelope	---	T12
Cathode	---	Coated filament

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8
Element	---	nc	f	---	2a	---	1a	---	f

ABSOLUTE RATINGS: See note 1

Parameter:	Ef	Epp/a	epx	ib	Io	Rl	CL	i(surge)	Alt
Unit:	Vac	Vac	v	ma	mAdc	Ohms	µF	a	ft
Maximum:	5.5	---	1,700	1,100	300	---	--	5.0	10,000
Minimum:	4.5	---	---	--	---	---	--	---	----
TEST CONDITIONS:	5.0	550	---	---	---	1,950	10	---	----

GENERAL:

Qualification - Required

Ⓑ denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						Min	Max	
(B)	Qualification test							
1031	Low-frequency vibration	No voltages	----	---	---	---	---	---
	Quality conformance inspection, part 1							
1353	Operation of rectifiers (1)	See notes 2 and 3	0.65	II	Io	285	---	mAdc
1231	Emission	Eb = 75 Vdc See note 4	0.65	II	Is	350	---	mAdc
	Quality conformance inspection, part 2							
----	Alinement	See note 5	----	---	---	---	---	---
1301	Filament current		----	---	If	2.7	3.3	A
1353	Operation of rectifiers (2)	Ef = 4.5 Vac	----	---	ΔIo If	---	10	%
1211	Insulation of electrodes		----	---	---	---	---	---
1101	Secureness of base, cap, or insert		----	---	---	---	---	---
1105	Permanence of marking		----	---	---	---	---	---
	Quality conformance inspection, part 3							
----	Life-test provisions	Group B, See note 6	----	---	---	---	---	---
----	Life-test end point (500 hours)							
1353	Operation of rectifiers (1)		----	---	Io	275	---	mAdc

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NOTES:

1. Tube shall be operated in a vertical position. Tube may be mounted in a horizontal position provided pins No. 2 and 4 are in a vertical plane.
2. In a full-wave circuit Zp/a shall be adjusted such that a tube having Etd = 44 Vdc at 225 mAdc per anode gives Io = 300 mAdc.
3. This test to be performed at the conclusion of the holding period.
4. Test each section separately. Anode voltage = 0 for section not under test.
5. The plane of the filament shall be parallel to the plane passing through the centerline of pins No. 2 and 4 plus or minus 22.5 degrees.
6. The values of R1 and CL given in test conditions may be considered as approximate, and shall be adjusted initially to give Io = 300 mAdc with ib equal to or greater than 1.1A.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Preparing activity:
Air Force - 85

Review activities:
Army - MU, MI
Air Force - 99
DSA - ES

(Project No. 5960-3127)

User activities:
Army - ME, WC
Navy - AS, OS, MC, CG, SH

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-E-1/985B Electron Tube, Receiving Type 5U4GB FSC 5960

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 OCT 76 **1426** PREVIOUS EDITION WILL BE USED.

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